

Math 514, F19
Quiz 6

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) How much do you need to invest at the beginning of each of the next 60 months in order to have a value of \$200,000 at the end of 60 months, given that the annual nominal interest rate is fixed at 6% and compounded monthly?

Monthly interest rate: $r = \frac{0.06}{12} = 0.005$

Suppose that we invest the amount of A per month, then

$$A + \frac{A}{(1+r)} + \frac{A}{(1+r)^2} + \dots + \frac{A}{(1+r)^{59}} = \frac{200000}{(1+r)^{60}}$$

$$A \left((1+r) + (1+r)^2 + \dots + (1+r)^{60} \right) = 200000$$

$$A \cdot \frac{(1+r) - (1+r)^{61}}{1 - (1+r)} = 200000$$

$$A = 200000 \cdot \left(\frac{-r}{1+r - (1+r)^{61}} \right) = 200000 \cdot \left(\frac{0.005}{1.00561 - 1.005} \right)$$

$$\approx 2852.3$$

Problem 2. (5 points) A ten-year \$10,000 bond costs \$10,000 and pays its holders \$500 every six months for ten years with a final additional payment of \$10,000 made at the end of those twenty payments. Assume the annual nominal interest rate is 6% and compounded continuously. Find the effective interest rate per six months and the present value of the bond.

The effective interest rate per six month is

$$r_{\text{eff}} = e^{\frac{0.06}{2}} - 1 \approx 0.03045$$

Thus

$$PV = -10000 + \sum_{i=1}^{20} \frac{500}{(1+r_{\text{eff}})^i} + \frac{10000}{(1+r_{\text{eff}})^{20}}$$

$$= -10000 + 500 \cdot \frac{1 - \frac{1}{(1+r_{\text{eff}})^{20}}}{1 - \frac{1}{1+r_{\text{eff}}}} + \frac{10000}{(1+r_{\text{eff}})^{20}}$$

$$= -10000 + 500 \cdot \frac{1 - \frac{1}{(1+r_{\text{eff}})^{20}}}{r_{\text{eff}}} + \frac{10000}{(1+r_{\text{eff}})^{20}}$$

$$\approx 2896.5$$